

The Power Criterion for One-Sided Drazin Invertibility

Automatic math-discovery run `fa_banach_001`

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Source question

Source: Kai Yan, *One-sided Drazin inverses in Banach algebras and perturbations of B-Fredholm spectra*, arXiv:2402.13574 (first posted February 21, 2024) [1].

Question 2.9 on page 8 asks whether a is left (respectively right) Drazin invertible with index n if and only if a^n is left (respectively right) group invertible.

Also

$$za^{n+1} = xa^{n-1}a^{n+1} = xa^{2n} = a^n.$$

Therefore $z = xa^{n-1}$ is the left Drazin inverse of a and the index of z is n . The proof of (ii) is dual to that of (i). $\square$

The aim of considering the relation between the one-side Drazin invertible element and its power is to give a one-sided version of [24, Corollary 5]. This is important because we try to use the spectral axiomatic theory of Mbekhta and Muller ([20]) to analysis the spectral properties of our one-sided Drazin invertible element in Banach algebras. However, the proof of [24, Corollary 5] depends highly on the commutativity of Drazin (or group) inverse. It seems difficult to give its one-sided analogues and so we list it as an open question.

Question 2.9. *Let $a \in \mathcal{A}$ and n be some non-negative integer. Is it true that a is left (resp. right) Drazin invertible with index n if and only if a^n is left (resp. right) group inverse?*

2.2 One-sided generalized Drazin inverses in Banach algebras

The Proposition 2.2 enables us to extend the definitions of left and right Drazin invert-

Result and convention

This packet gives a **candidate full resolution**. For every positive integer n , the power condition is equivalent to one-sided Drazin index *at most* n . Equivalently, it is equivalent to the existence of a one-sided Drazin inverse satisfying the defining equations with exponent n . The proof is purely algebraic and works in every unital ring.

This distinction resolves two wording issues in the printed question. If “index” means the minimal index—as the author explicitly specifies in a later paper [2]—then equality with n is too strong: for example, $a = 1$ has minimal index zero although $a^n = 1$ is group invertible for every $n \geq 1$. Also, the printed allowance $n = 0$ is false in general because $a^0 = 1$ is always group invertible, whereas not every a is one-sided invertible (take $a = 0$).

Definitions

Let $\mathcal{R}$ be a unital ring. An element $b \in \mathcal{R}$ is *left group invertible* if some $x \in \mathcal{R}$ satisfies

$$bxb = xb^2, \quad x^2b = x, \quad xb^2 = b.$$

An element $a \in \mathcal{R}$ admits a *left Drazin inverse with exponent n* if some $z \in \mathcal{R}$ satisfies

$$aza = za^2, \quad z^2a = z, \quad za^{n+1} = a^n. \tag{1}$$

Its minimal possible exponent is denoted $\text{ind}_L(a)$. The right-hand notions reverse all products in these equations.

Idea of the proof

Suppose $b = a^n$ has a left group inverse x . The idempotent $e = xb$ splits the ring into Peirce corners. With $p = 1 - e$, the element a is lower triangular:

$$a = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}, \quad A^n \text{ is left invertible}, \quad D^n = 0.$$

Thus A itself has a left inverse L . The finite Sylvester sum

$$T = \sum_{k=0}^{n-1} D^k C L^{k+1}$$

satisfies $TA - DT = C$. Since $T^2 = 0$, conjugation by $1 + T$ removes the off-diagonal corner and turns a into $A + D$. The element L , placed in the A -corner, is visibly a left Drazin inverse with exponent n .

Sharp power theorem

Theorem 1. *Let $\mathcal{R}$ be a unital ring, let $a \in \mathcal{R}$, and let $n \geq 1$. The following are equivalent:*

1. a^n is left group invertible;
2. a admits a left Drazin inverse satisfying (1) with exponent n ;
3. a is left Drazin invertible and $\text{ind}_L(a) \leq n$.

The analogous three right-handed statements are equivalent.

Proof. We prove the nontrivial implication (1) $\Rightarrow$ (2). Put $b = a^n$, and let x be a left group inverse of b . Its defining equations give

$$bxb = b, \quad x^2b = x, \quad xb^2 = b.$$

Set $e = xb$ and $p = 1 - e$. Then

$$e^2 = x(bxb) = xb = e, \quad be = b = eb, \quad bp = pb = 0.$$

The right annihilator of b is exactly $p\mathcal{R}$: if $br = 0$, then $er = xbr = 0$, hence $r = pr$; the reverse inclusion follows from $bp = 0$. Since a commutes with $b = a^n$, we have $b(ap) = a(bp) = 0$, and therefore $ap = pap$, or $eap = 0$.

Define the Peirce corners

$$A = eae \in e\mathcal{R}e, \quad C = pae \in p\mathcal{R}e, \quad D = pap \in p\mathcal{R}p.$$

Thus $a = A + C + D$, with the formal block shape displayed in the proof idea. Taking powers and using $bp = pb = 0$ gives

$$A^n = ea^n e = b, \quad D^n = pa^n p = 0. \tag{2}$$

Moreover $h = exe$ is a left inverse of A^n in $e\mathcal{R}e$, since

$$hA^n = exeb = exb = e.$$

Consequently

$$L = hA^{n-1} \in e\mathcal{R}e, \quad LA = e.$$

Now put

$$T = \sum_{k=0}^{n-1} D^k C L^{k+1} \in p\mathcal{R}e. \quad (3)$$

Because $LA = e$ and $D^n = 0$, the sum telescopes:

$$TA - DT = \sum_{k=0}^{n-1} D^k C L^k - \sum_{k=1}^n D^k C L^k = C. \quad (4)$$

Also $T^2 = 0$, so $u = 1 + T$ is invertible with $u^{-1} = 1 - T$. Direct Peirce multiplication, using (4), yields

$$u^{-1}au = A + D. \quad (5)$$

In the diagonalized coordinates set $z' = L$. Orthogonality of the two corners, $LA = e$, and $D^n = 0$ give

$$\begin{aligned} (A + D)L(A + D) &= ALA = LA^2, \\ L^2(A + D) &= L, \\ L(A + D)^{n+1} &= LA^{n+1} = A^n = (A + D)^n. \end{aligned}$$

Therefore $z = uLu^{-1}$ satisfies all three equations (1) for a , proving (2).

For (2) $\Rightarrow$ (1), let z satisfy (1) and put $e = za$, $p = 1 - e$. The same elementary identities used in the source's Proposition 2.7 show that p commutes with a , $(ap)^n = 0$, and the e -corner of a is left invertible. Hence the e -corner of a^n is left invertible and its p -corner is zero. If H is a left inverse of that e -corner, then, extended by zero on the p -corner,

$$(a^n)H(a^n) = H(a^n)^2 = a^n, \quad H^2a^n = H;$$

so H is a left group inverse of a^n .

Finally, (2) and (3) are equivalent because an equation $za^{k+1} = a^k$ with $k \leq n$ remains true after right multiplication by a^{n-k} . This proves the left-handed result. Applying it in the opposite ring $\mathcal{R}^{\text{op}}$ proves the right-handed result. $\square$

Corollary 1 (Exact minimal index). *For $n \geq 2$, $\text{ind}_L(a) = n$ if and only if a^n is left group invertible but a^{n-1} is not. For $n = 1$, exact index one means that a is left group invertible but not left invertible. The right-handed statements are dual.*

Machine-checkable audit

The accompanying exact SymPy script instantiates a nontrivial 5×5 lower-triangular example with $n = 3$. It verifies the Sylvester identity, the similarity (5), the commuting idempotent, nilpotence of the bad corner, all three constructed left-Drazin equations, and all three left-group equations for a^n . No floating-point computation is used.

Novelty, limitations, and review recommendation

A bounded search of the run indexes, exact question wording, core keywords, and the author's 2025 follow-up [2] found no later source proving this power criterion. The follow-up clarifies the minimal-index convention but studies other fundamental properties. Novelty confidence is moderate, pending specialist review.

The result concerns ordinary one-sided Drazin invertibility, not the generalized Drazin notions later in the source paper. It strengthens the ambient category from Banach algebras to unital rings. Promotion for expert review is recommended; the main review points are the annihilator identity $\text{ann}_r(b) = p\mathcal{R}$ and the finite Sylvester conjugation (3)–(5).

References

- [1] K. Yan, *One-sided Drazin inverses in Banach algebras and perturbations of B-Fredholm spectra*, arXiv:2402.13574.
- [2] K. Yan, *On the spectral identities and fundamental properties of one-sided Drazin inverses in Banach algebras*, arXiv:2504.18995.